

Q1 2021 (01 Sep Shift 2)

Let $[t]$ denote the greatest integer $\leq t$. The number of points where the function

$$f(x) = [x] |x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2, 2)$$

is not continuous is .

Q2 2021 (31 Aug Shift 1)

$$\text{If the function } f(x) = \begin{cases} \frac{1}{x} \log_e \left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}} \right), & x < 0 \\ k, & x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1}-1}, & x > 0 \end{cases}$$

is continuous at $x = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to :

(1) -5

(2) 5

(3) -4

(4) 4

Q3 2021 (31 Aug Shift 1)

The function $f(x) = |x^2 - 2x - 3| \cdot e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly:

(1) four points

(2) three points

(3) two points

(4) one point

Q4 2021 (26 Aug Shift 2)

Let $[t]$ denote the greatest integer less than or equal to t . Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$, and

$h(x) = \min\{f(x), g(x)\}$, $x \in [-2, 2]$. Then h is :

(1) continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$

(2) not continuous at exactly three points in $[-2, 2]$

(3) continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$

(4) not continuous at exactly four points in $[-2, 2]$

Q5 2021 (26 Aug Shift 1)

Let $a, b \in \mathbf{R}$, $b \neq 0$, Define a function

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1), & \text{for } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & \text{for } x > 0 \end{cases}$$

If f is continuous at $x = 0$, then $10 - ab$ is equal to .

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Answer Key

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Q1 (2)

Q2 (1)

Q3 (3)

Q4 (1)

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Q5 (14)

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Q1 (2)

$$f(x) = [x] |x^2 - 1| + \sin \frac{\pi}{[x+3]} - [x+1]$$

$$f(x) = \begin{cases} 3 - 2x^2, & -2 < x < -1 \\ x^2, & -1 \leq x < 0 \\ \frac{\sqrt{3}}{2} + 1, & 0 \leq x < 1 \\ x^2 + 1 + \frac{1}{\sqrt{2}}, & 1 \leq x < 2 \end{cases}$$

discontinuous at $x = 0, 1$

Q2 (1)

If $f(x)$ is continuous at $x = 0$, $RHL = LHL = f(0)$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2 + 1} - 1} \cdot \frac{\sqrt{x^2 + 1} + 1}{\sqrt{x^2 + 1} + 1} \quad (\text{Rationalisation})$$

$$\lim_{x \rightarrow 0^+} -\frac{2 \sin^2 x}{x^2} \cdot (\sqrt{x^2 + 1} + 1) = -4$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x} \ell \left(\frac{1 + \frac{x}{a}}{1 - \frac{x}{b}} \right)$$

$$\lim_{x \rightarrow 0} \frac{\ell n(1 + \frac{x}{a})}{(\frac{x}{a}) \cdot a} + \frac{\ell n(1 - \frac{x}{b})}{(-\frac{x}{b}) \cdot b}$$

$$= \frac{1}{a} + \frac{1}{b}$$

$$\text{So } \frac{1}{a} + \frac{1}{b} = -4 = k$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{4}{k} = -4 - 1 = -5$$

Q3 (3)

$$f(x) = |(x-3)(x+1)| \cdot e^{(3x-2)^2}$$

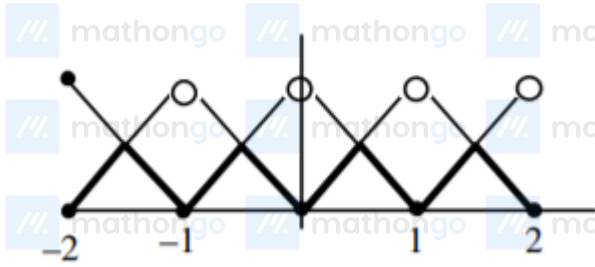
$$f(x) = \begin{cases} (x-3)(x+1) \cdot e^{(3x-2)^2} & ; x \in (3, \infty) \\ -(x-3)(x+1) \cdot e^{(3x-2)^2} & ; x \in [-1, 3] \\ (x-3) \cdot (x+1) \cdot e^{(3x-2)^2} & ; x \in (-\infty, -1) \end{cases}$$

Clearly, non-differentiable at $x = -1$ & $x = 3$.

Q4 (1)

$$\min\{x - [x], 1 - x + [x]\}$$

$$h(x) = \min\{x - [x], 1 - [x - [x]]\}$$



$\Rightarrow$ always continuous in $[-2, 2]$

but non differentiable at 7 Points

Q5 (14)

$$f(x) = \begin{cases} a \sin \frac{\pi}{2}(x-1), & x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & x > 0 \end{cases}$$

For continuity at '0'

$$\lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\tan 2x - \sin 2x}{bx^3} = -a$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\frac{8x^3}{3} + \frac{8x^3}{3!}}{bx^3} = -a$$

$$\Rightarrow 8 \left(\frac{1}{3} + \frac{1}{3!} \right) = -ab$$

$$\Rightarrow 4 = -ab$$

$$\Rightarrow 10 - ab = 14$$